



Smart Air Purifier

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Introduction

This presentation showcases a smart air purifier project utilizing Raspberry Pi 5 for real-time air quality monitoring and purification. The system employs a range of sensors to detect pollution levels, controls a fan-based purifier accordingly, and provides data visualization and remote monitoring capabilities.

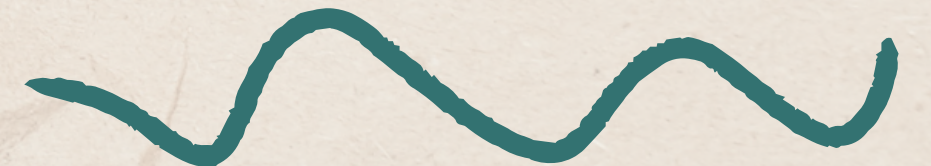


Project Goals

The Smart Air Purifier aims to monitor air quality in real time using PM2.5 and DHT11 sensors.

It automatically activates a fan when pollution levels rise, ensuring clean and breathable indoor air.

The system is IoT-enabled via Blynk, allowing users to monitor and control it remotely through their smartphone.





Problems

Indoor air pollution poses serious health risks, yet most purifiers lack automation and remote monitoring. There's a need for a smart, affordable system that can detect poor air quality and respond in real time.





Process

1

Sensing

The DHT11 and PM2.5 sensors continuously monitor temperature, humidity, and air quality levels.

2

Processing

The Raspberry Pi 5 analyzes sensor data and decides when to activate the air purifier (fan).

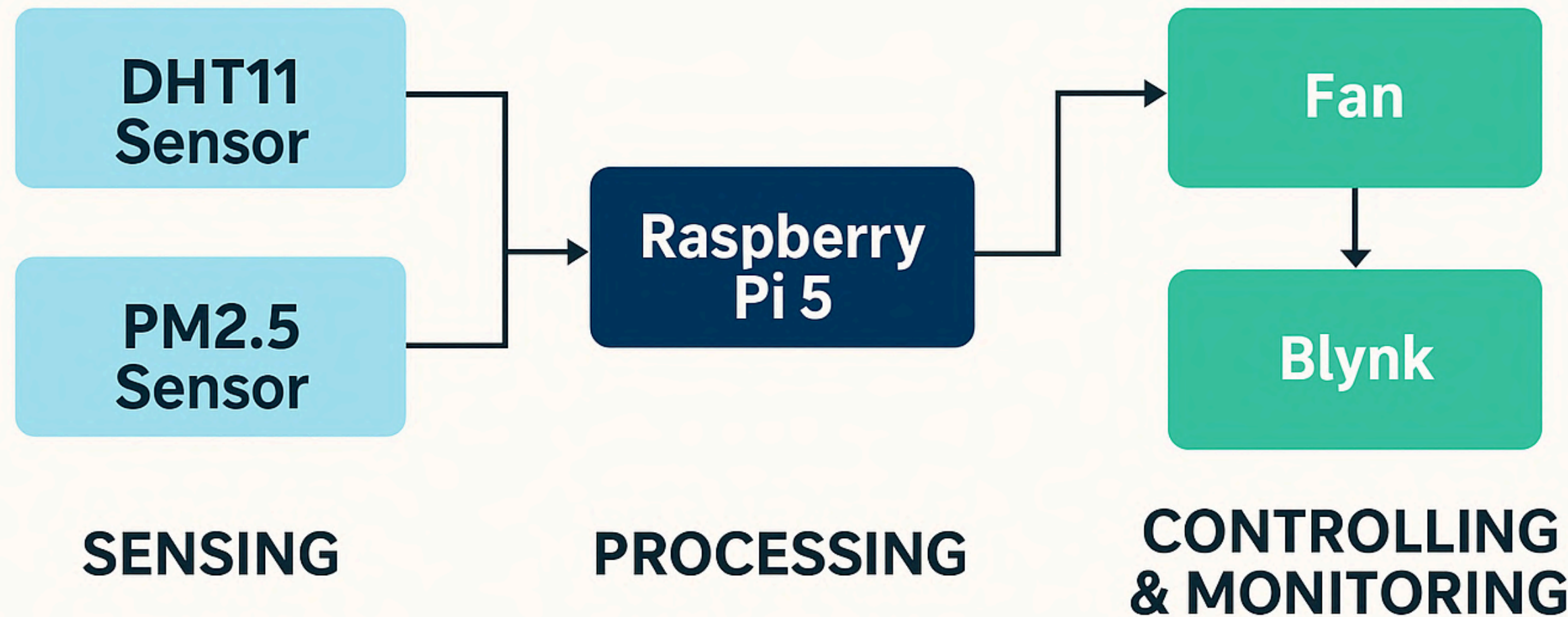
3

Controlling & Monitoring

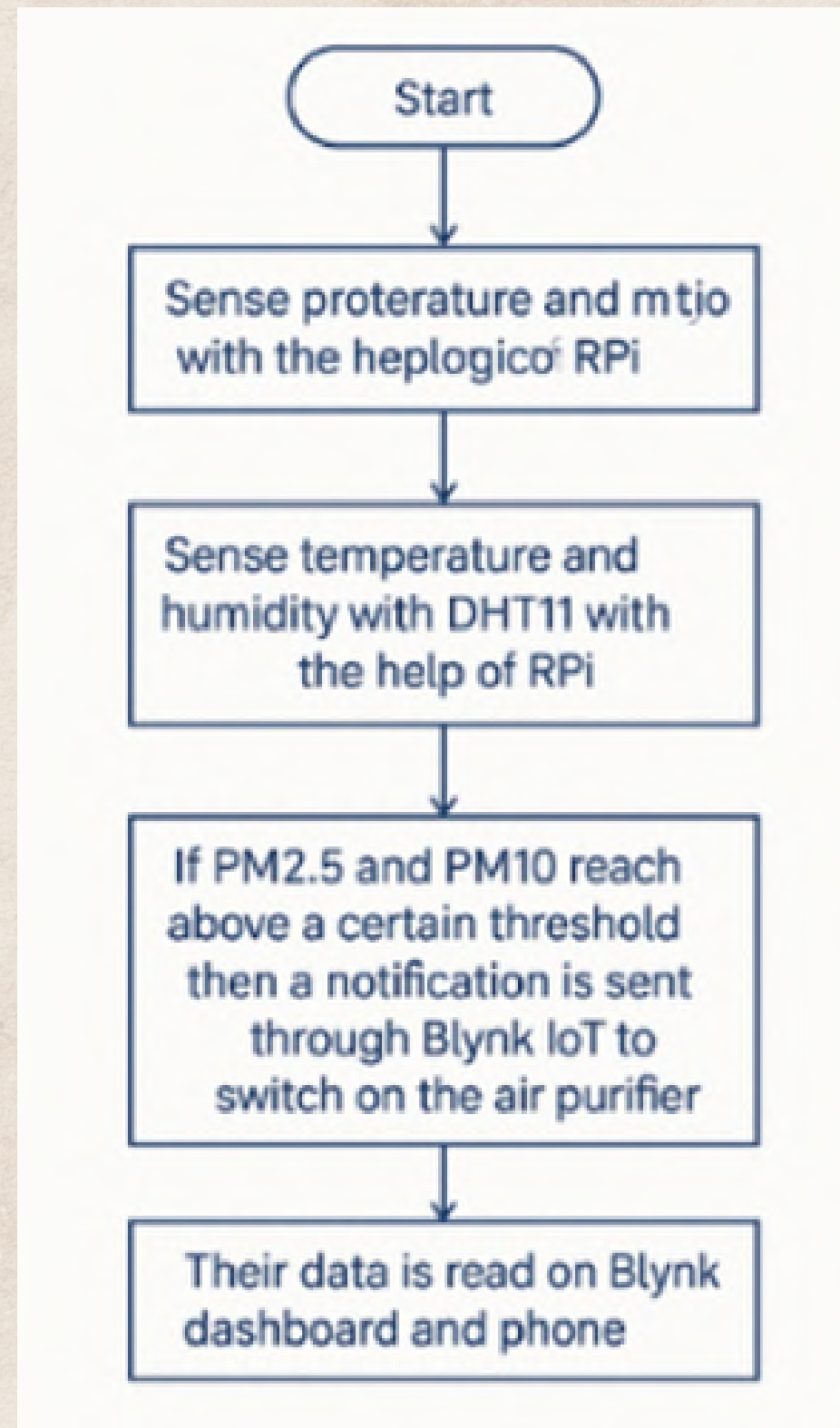
The fan is turned ON/OFF based on pollution levels, and real-time data is sent to the Blynk app for remote monitoring.

Block Diagram

SMART AIR PURIFIER SYSTEM



Flow Chart





Conclusion

The smart air purifier is an innovative solution for real-time air quality monitoring and improvement. By utilizing various sensors and the Raspberry Pi 5, the system effectively responds to changing environmental conditions, providing a cost-effective and efficient method for maintaining healthier air indoors. Future enhancements could include the integration of AI technologies for predictive analysis and broader IoT capabilities.



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Thank
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